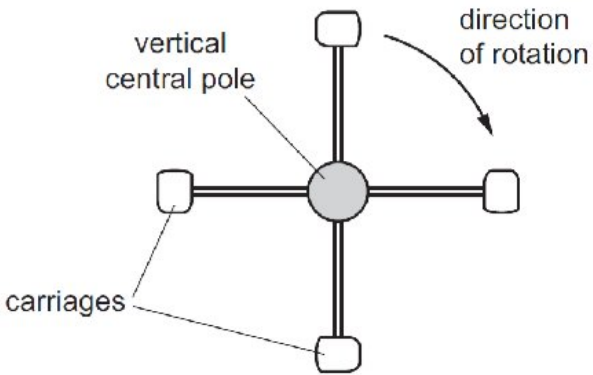
6 A fairground ride consists of four carriages connected to a vertical central pole, as shown in the following view from the top of the ride.



not to
scale

A motor rotates the central pole about its axis. This results in the four carriages each moving along a circular path.

The distance from the middle of each carriage to the centre of the pole is 3.20 m. When they are moving, each carriage experiences an air resistance force of 85.0 N. Assume that there are no other significant resistive forces.

Which torque does the motor need to apply to the pole to keep the system rotating at constant maximum speed?

A 5.44 N m **B** 272 N m **C** 544 N m **D** 1090 N m